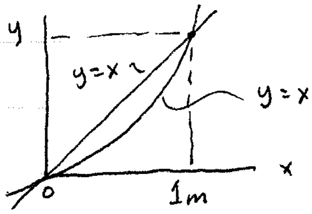
BE 50 HW

Area Moment of Inertia by integration



For the area, find I_x, I_y, I_{xy}, J_o, k_x, k_y, $\bar{x}$, $\bar{y}$, A

① Area: $A = \int_A dA = \int_0^1 \int_{x^2}^{x} dy\,dx = \int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \boxed{\frac{1}{6}\ m^2 = \text{Area}}$ .167

Centroids

② $\bar{y}$: $\bar{y} = \dfrac{\int_A \tilde{y}\,dA}{\int_A dA} = \dfrac{\int_0^1 \int_{x^2}^{x} y\,dy\,dx}{1/6} = \dfrac{\int_0^1 \frac{1}{2}(x^2 - x^4)\,dx}{1/6} = \dfrac{\frac{1}{6} - \frac{1}{10}}{1/6} = \frac{12}{30} = .4m$ $\boxed{\bar{y} = .4m}$

③ $\bar{x} = \dfrac{\int_A \tilde{x}\,dA}{\int_A dA} = \dfrac{\int_0^1 \int_{x^2}^{x} x\,dy\,dx}{1/6} = \dfrac{\int_0^1 (x^2 - x^3)\,dx}{1/6} = \dfrac{1/3 - 1/4}{1/6} = \frac{6}{12} = \boxed{.5m = \bar{x}}$

Area moment of Inertia (MOI)

④ $I_x = \int_A y^2\,dA = \int_0^1 \int_{x^2}^{x} y^2\,dy\,dx = \int_0^1 \frac{1}{3}(x^3 - x^6)\,dx = \frac{1}{3}\left(\frac{1}{4} - \frac{1}{7}\right) = \boxed{\frac{1}{28}\ m^4 = I_x}$ .0357

⑤ $I_y = \int_A x^2\,dA = \int_0^1 \int_{x^2}^{x} x^2\,dy\,dx = \int_0^1 (x^3 - x^4)\,dx = \frac{1}{4} - \frac{1}{5} = \boxed{\frac{1}{20}\ m^4 = I_y}$ .050

Polar MOI

⑥ $J_o = I_x + I_y = \frac{1}{28} + \frac{1}{20} = \frac{12}{140} = \frac{3}{35} = \boxed{0.0857\ m^4 = J_o}$

Product of Inertia

⑦ $I_{xy} = \int_A xy\,dA = \int_0^1 \int_{x^2}^{x} xy\,dy\,dx = \int_0^1 \frac{1}{2}(x^3 - x^5)\,dx = \frac{1}{2}\left(\frac{1}{4} - \frac{1}{6}\right) = \boxed{\frac{1}{24}\ m^4 = I_{xy}}$ .0417

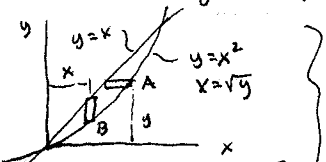
Radius of Gyration

⑧ $k_x = \sqrt{I_x / A} = \left[\dfrac{1/28}{1/6}\right]^{1/2} = \left[6/28\right]^{1/2} = \left[3/14\right]^{1/2} = \boxed{0.4629\ m = k_x}$

⑨ $k_y = \sqrt{I_y / A} = \sqrt{\dfrac{1/20}{1/6}} = \sqrt{6/20} = \sqrt{3/10} = \boxed{0.5477\ m = k_y}$

⑩ $k_o = \sqrt{J_o / A} = \sqrt{\dfrac{3/35}{1/6}} = \sqrt{\dfrac{18}{35}} = 0.7171\ m$

Use of a single integral:



$dI = d\bar{I} + dA\,d^2$

Element A:

$I_x = \int dI_x = \int_0^1 y^2(\sqrt{y} - y)\,dy = \int_0^1 (y^{5/2} - y^3)\,dy = \frac{2}{7} - \frac{1}{4} = \boxed{\frac{1}{28}\ m^4 = I_x}$

$I_y = \int_0 dI_y = \int_0^1 x^2(x - x^2)\,dx = \int_0^1 x^3 - x^4\,dx = \frac{1}{4} - \frac{1}{5} = \boxed{\frac{1}{20}\ m^4 = I_y}$

Element B: